Machine Learning: Midterm

Solutions

1. Evaluating classifiers (20 points)

A university wants to use an AI-text detector to check student essays. To test it, the detector is run on 400 essays whose authorship is known: 60 were written with AI and 340 by students. Of the 400 essays, the detector flags 50 as written with AI. Of these 50 flagged essays, 45 really were written with AI. Treat "written with AI" as the positive class.

1.1 [4 points] Write down the confusion matrix.

Solution: 60 AI-written essays and 340 human-written essays.

	flagged	not flagged
written with AI	TP = 45	FN = 15
written by students	FP = 5	TN = 335

1.2 [2 points] What are precision and recall for this detector?

Solution: Precision = $45/50 = 0.90$. Recall = $45/60 = 0.75$.

1.3 [2 points] Do we care more about improving precision or recall for this problem, and why?

Solution: Precision. A false positive accuses an innocent student of cheating; a missed AI essay (false negative) is a smaller harm. A flag should almost always be right before anyone acts on it.

The detector also gives each essay a score between 0 and 1; a higher score means "more likely written with AI". Here are 8 essays, 4 written with AI and 4 by students:

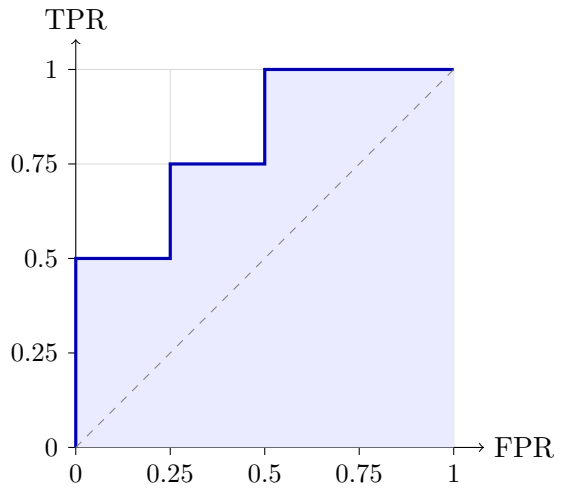
essay	1	2	3	4	5	6	7	8
score	0.95	0.85	0.70	0.60	0.50	0.40	0.30	0.10
written with AI?	yes	yes	no	yes	no	yes	no	no

We say the detector flags an essay as AI-written if its score is over t.

1.4 [10 points] Draw the ROC curve.

Solution: As t passes each essay, from the highest score down, an AI essay moves the point up by $1/4 = 0.25$ and a human essay moves it right by 0.25. Starting at $(0, 0)$, the points are

$$(0, 0.25),\ (0, 0.5),\ (0.25, 0.5),\ (0.25, 0.75),\ (0.5, 0.75),\ (0.5, 1),\ (0.75, 1),\ (1, 1).$$



1.5 [2 points] Compute the AUC.

Solution: The area under the staircase, in columns of width 0.25 with heights $0.5, 0.75, 1, 1$: $\text{AUC} = 0.25 \times (0.5 + 0.75 + 1 + 1) = 0.8125 \approx 0.81$.

2. Support vector machines (20 points)

2.1 [10 points] A soft-margin SVM returned $\boldsymbol{w} = (6, 8)$, $b = -20$.

point	A	B	C	D	E
x_1	3	1	1.5	1.25	3
x_2	1	1	1.5	1.5	0.5
label	+	−	+	−	−

(a) Compute $f(\boldsymbol{x})$ and the predicted class of each point.

(b) What is the width of the margin?

(c) Compute each point's distance to the decision boundary.

(d) Which points are support vectors? Which point is on the margin, which is inside it, and which is misclassified?

Solution: $f(\boldsymbol{x}) = 6x_1 + 8x_2 - 20$ and $\|\boldsymbol{w}\| = \sqrt{6^2 + 8^2} = 10$.

point	A	B	C	D	E
$f(\boldsymbol{x})$	6	−6	1	−0.5	2
predicted class	+	−	+	−	+
distance $\lvert f(\boldsymbol{x})\rvert/\|\boldsymbol{w}\|$	0.6	0.6	0.1	0.05	0.2
$y\, f(\boldsymbol{x})$	6	6	1	0.5	−2

(b) Margin width $= 2/\|\boldsymbol{w}\| = 2/10 = 0.2$.

(d) The support vectors are the points with $y\, f(\boldsymbol{x}) \le 1$: C, D and E. C is on the margin ($y\, f = 1$), D is inside the margin but correctly classified ($y\, f = 0.5$), and E is misclassified (label −, predicted +).

2.2 [4 points] The soft-margin SVM solves

$$\min_{\boldsymbol{w}, b, \boldsymbol{\xi}} \ \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C \sum_i \xi_i \quad \text{subject to} \quad y_i(\boldsymbol{w} \cdot \boldsymbol{x}_i + b) \ge 1 - \xi_i, \ \ \xi_i \ge 0.$$

(a) What does the slack ξ_i measure, and why do we need it?

(b) Where does a training point lie, relative to the margin and the decision boundary, if $\xi_i = 0$? If $0 < \xi_i < 1$? If $\xi_i > 1$?

Solution: (a) ξ_i measures how far point i falls short of being on its correct side, outside the margin. Without slack, the constraints $y_i(\boldsymbol{w} \cdot \boldsymbol{x}_i + b) \geq 1$ have no solution when the classes overlap; slack lets some points violate the margin, at a cost $C\xi_i$ each.

(b) $\xi_i = 0$: on its correct side, on or outside the margin. $0 < \xi_i < 1$: inside the margin, but still correctly classified. $\xi_i > 1$: on the wrong side of the decision boundary (misclassified). (At the solution $\xi_i = \max(0, 1 - y_i f(\boldsymbol{x}_i))$, so these are the cases $y_i f \geq 1$, $0 < y_i f < 1$ and $y_i f < 0$.)

2.3 [4 points] Two soft-margin SVMs are trained on the same data, one with $C = 0.01$ and one with $C = 100$.

(a) Which has the wider margin, and why?

(b) Which has more support vectors, and why?

(c) Which is more likely to overfit, and why?

(d) How would you choose C in practice, and why?

Solution: (a) $C = 0.01$: slack is cheap, so the SVM accepts many points inside the margin in exchange for a wider one. (b) $C = 0.01$: with a wider margin, more points fall on or inside it, and all of them are support vectors. (c) $C = 100$: violations are expensive, so the margin narrows to fit individual training points. (d) C is a hyperparameter: try several values and pick the one with the best accuracy on validation data (or by cross-validation). Training accuracy cannot choose it, because a larger C always fits the training data at least as well; only held-out data shows which C generalizes.

2.4 [2 points] You remove a training point that is not a support vector and retrain the SVM. What changes? Why?

Solution: Nothing. The solution depends only on the support vectors; a point outside the margin does not affect $\boldsymbol{w}$ or b.

3. Nonlinear regression (20 points)

3.1 [4 points] kNN regression is trained on six points:

x	1	2	4	5	7	8
y	3	6	4	8	10	12

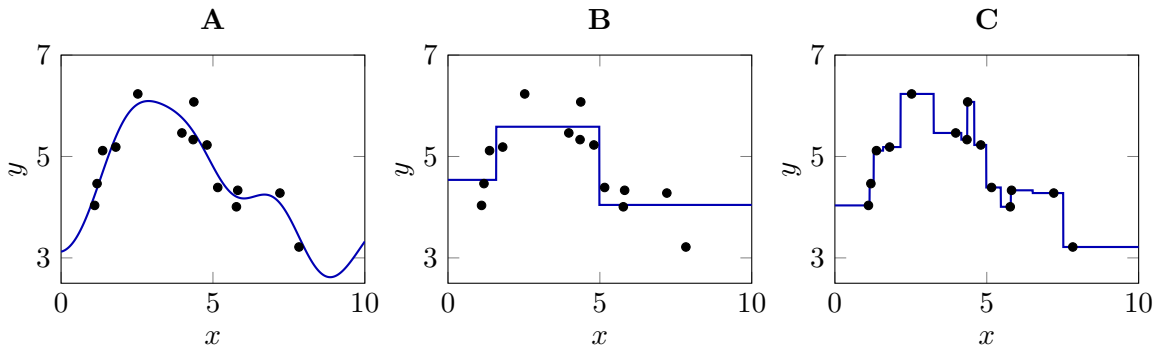
Predict y at $x = 4.4$ using (a) $k = 1$ and (b) $k = 3$.

Solution: (a) The nearest point is $x = 4$, so the prediction is 4. (b) The three nearest points are $x = 4, 5, 2$ (distances $0.4, 0.6, 2.4$), so the prediction is $(4 + 8 + 6)/3 = 6$.

3.2 [6 points] For each of a regression tree, kNN regression and SVR, name a hyperparameter that controls how closely the model follows the training data. Which setting of it makes overfitting more likely?

Solution: Regression tree: a larger maximum depth, or a smaller minimum number of points needed to split. kNN: a smaller k. SVR: a larger C, or a smaller ε.

3.3 [5 points] The figure shows three models fitted to the same data. Which panel shows a regression tree, which shows kNN with $k = 1$, and which shows SVR with a kernel? Explain each.



Solution: A is SVR with a kernel: a smooth curve. B is the regression tree: a few wide flat steps, one for each leaf, which predicts the average y of its points. C is kNN with $k = 1$: a narrow step for every training point, passing through every point.

3.4 [5 points] We use kNN to predict house prices from floor area (500 to 5,000 square feet) and number of bedrooms (1 to 6). Why might kNN do badly on these raw features? What would you do?

Solution: Distances are dominated by floor area: a difference of one bedroom counts as much as one square foot, so the nearest houses match on area and almost ignore bedrooms. Standardize each feature (subtract its mean, divide by its standard deviation) before computing distances.